Check for updates

Received: 8 November 2023 | Revised: 5 December 2023 | Accepted: 11 December 2023

DOI: 10.1002/dta.3631

RESEARCH ARTICLE

WILEY

Prevalence of carboxy-Δ^8-tetrahydrocannabiniol in antidoping samples

Vinod S. Nair[1] | Mari Heybroek[1] | Emily Boyle[1] | Mason Rogers[2] | Thane Campbell[1] | Daniel Eichner[1] | Kevin Hill[3]

[1]Sports Medicine Research and Testing Laboratory, South Jordan, Utah, USA

[2]University of Utah School of Medicine, Salt Lake City, Utah, USA

[3]Harvard Medical School, Boston, Massachusetts, USA

Correspondence
Vinod Nair, Sports Medicine Research and Testing Laboratory, 10644 South Jordan Gateway, South Jordan, UT 84095, USA.
Email: vnair@smrtl.org

Funding information
Sports Medicine Research and Testing Laboratory

Abstract

Δ^9-Tetrahydrocannabinol (Δ^9-THC) is usually the primary psychoactive agent in cannabis preparations. Recently, products containing another isomer, Δ^8-tetrahydrocannabinol (Δ^8-THC), have become available for sale. Δ^8-THC exists naturally in the cannabis plant at very low concentrations; hence, the Δ^8-THC present in most of the above-mentioned products is likely to be manufactured synthetically. A surge in popularity of these products, coupled with little oversight to ensure purity and potency, has led to reports of adverse events. Workplace drug testing programs as well as many sporting organizations prohibit the use of cannabinoids. Carboxy-Δ^9-THC (Δ^9-THC-COOH) is the targeted urinary metabolite for detection of cannabis use. The proliferation of products containing Δ^8-THC, which metabolizes to Δ^8-THC-COOH, presents analytical complexity with respect to separation and quantification of the individual isomers as well as legal complexity with respect to lack of clarity around the legal status of Δ^8-THC. This study aims to estimate the prevalence of Δ^8-THC use in the athlete community by monitoring for Δ^8-THC-COOH in samples collected for antidoping. A high-performance liquid chromatography tandem mass spectrometry (LC–MS/MS) method was utilized to resolve Δ^8 and Δ^9-THC-COOH. One thousand samples with a presumptive Δ^9-THC-COOH finding in routine screening were analyzed by the above LC–MS/MS method. Approximately 12% of samples contained Δ^8-THC-COOH at relative abundances between 5% and 100% of total carboxy-THC content.

KEYWORDS
synthetic cannabinoids, Δ^8-THC, Δ^9-THC

1 | INTRODUCTION

Tetrahydrocannabinol (THC), which is one of almost 150 cannabinoids in the cannabis plant, is primarily responsible for the psychoactive properties of marijuana.[1–3] The term THC usually refers to Δ^9-tetrahydrocannabinol (Δ^9-THC). Other isomers, such as Δ^8-THC, do exist naturally but in very small quantities relative to the Δ^9 isomer.[4] Since the passage of the Farm Bill of 2018, a number of products advertised as primarily containing Δ^8-THC have become available for sale including vapes, tinctures, gummies, and infused beverages, all of which are used in a similar manner as Δ^9-THC products.[5] The Farm Bill effectively legalized the sale of hemp-derived Δ^8-THC products with little oversight. In a recent online survey, most Δ^8-THC users described relaxation, euphoria, and pain relief along with a moderate amount of cognitive distortions such as difficulty concentrating, difficulty with short-term memory, and an altered sense of time.[6] Users generally compared Δ^8-THC favorably with both Δ^9-THC and pharmaceutical drugs and therefore reported substituting Δ^8-THC

 © 2024 John Wiley & Sons Ltd.

products for these. This surge in popularity has created a significant new public health issue.

Due to its low natural abundance, the Δ^8-THC in these products is likely of synthetic origin. Indeed, it has been shown that cannabidiol (CBD)—a non-psychoactive cannabinoid present in cannabis—can be converted to Δ^8-THC by treatment with acid and heat.[7] It is plausible that this synthetic process produces harmful byproducts that have not been characterized and improper purification may result in these byproducts being present in the finished product offered for sale.[8] The US Food and Drug Administration (FDA) reported 104 adverse events in patients who consumed Δ^8-THC products between December 1, 2020, and February 28, 2022, which may be related to contaminants introduced by the chemical synthesis process.[9] Consequently, the US FDA and the Centers for Disease Control and Prevention (CDC) have issued warnings about the health risks of products containing Δ^8-THC.[9,10]

From a legal perspective, cannabis strains that are high in CBD but contain <0.2% Δ^9-THC (European Union laws) or <0.3% Δ^9-THC (USA/Canada laws) are considered legal. If CBD derived from these sources is used to manufacture Δ^8-THC, the legal status of the resulting Δ^8-THC is unclear and can potentially present medico-legal complexities.[11] Further, the synthetic process to manufacture Δ^8-THC is likely to produce both Δ^8 and Δ^9 isomers, and inadequate post-synthesis purification can result in both isomers being present in the final product.[7]

Currently, workplace drug testing programs as well as sporting federations specify detection of carboxy-Δ^9-tetrahydrocannabinol (Δ^9-THC-COOH) for detection of prohibited cannabis use. Some athletes, seeking the effects of cannabis while knowing they will be tested for Δ^9-THC, may opt to use Δ^8-THC instead to minimize the likelihood of a positive drug test. The status of Δ^8-THC has not specifically been addressed to date. Further, it is possible that the assay for quantitation of Δ^9-THC-COOH may overestimate the concentration if significant amounts of the Δ^8 isomer are present unless the method resolves the two isomers. Indeed, the potential for such an occurrence was presented by Rzeppa et al., where contradictory results were obtained between the screening and confirmation quantitation for Δ^9-THC-COOH due to the presence of the Δ^8 isomer.[12] The object of this study was to investigate the use of Δ^8-THC in the athlete population by the detection of Δ^8-THC-COOH in samples collected for anti-doping purposes and evaluate the potential for overestimation of the threshold substance Δ^9-THC-COOH.

2 | MATERIAL AND METHODS

2.1 | Reagents and chemicals

N-Trimethylsilyl-*N*-methyl trifluoroacetamide (MSTFA) was obtained from Machery Nagel (Duren, Germany), ammonium iodide from Sigma Aldrich (St Louis, MO, USA), and ethanethiol from Alfa Aesar (Merelbeke, Belgium). Methanol, acetonitrile, and methyl-t-butyl ether were purchased from Honeywell (Muskegon, MI, USA). β-Glucuronidase was obtained from Roche Diagnostics (Mannheim, Germany). Δ^9-THC-COOH and d3-Δ^9-THC-COOH were obtained from Cerilliant (Round Rock, TX, USA) while Δ^8-THC-COOH was obtained from Cayman Chemical (Ann Arbor, MI, USA).

2.2 | Samples

Routine anti-doping samples that showed a presumptive finding for THC-COOH above 5 ng/mL in the initial screening procedure were investigated for this study. The population comprised a mix of samples from Olympic sports (in and out of competition) as well as professional sports.

2.3 | Sample preparation

All samples were processed using routine methodology established in the laboratory for antidoping testing. In brief, 2.5 mL of urine was subjected to enzymatic hydrolysis and extracted with 4 mL MTBE. A portion of the extract (1.6 mL) was dried and derivatized with activated MSTFA for GC–MS/MS analysis while 0.8 mL of the extract was dried and reconstituted with DMSO:0.1% formic acid for LC–MS/MS analysis.[13] When a potential signal for THC-COOH above 5 ng/mL was detected in the GC–MS/MS screen, the corresponding LC–MS/MS extract was re-injected on a confirmation method to evaluate whether Δ^8-THC-COOH was also present.

2.4 | LC–MS/MS confirmation

A validated and routinely used Δ^9-THC-COOH confirmation method was modified slightly to improve resolution between the Δ^9 and Δ^8 isomers. The analysis was performed on an Acquity UPLC coupled to a Waters TQ-XS mass spectrometer (Waters Corporation, Milford,

TABLE 1 LC conditions (top panel) and MRM transitions (bottom panel) used in the method to resolve the two THC-COOH isomers.

Time	Flow (mL/min)	%A (0.1% formic acid)	%B (acetonitrile)
0	0.45	70	30
7.3	0.45	42	58
8	0.45	5	95
8.1	0.45	70	30
9	0.45	70	30

Compound	Precursor (*m/z*)	Daughter (*m/z*)	Cone (V)	Collision (eV)
THC-COOH	343.2	299	20	20
		245.2	20	28
		325.2	20	20
Δ^9-THC-COOH d3	346.2	302	20	20
		248.1	20	28

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3631 by Ohio State University Ohio Sta, Wiley Online Library on [16/01/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

MA, USA). Chromatographic separation was achieved with a BEH C18 column (2.1 × 50 mm, 1.7 µm; Waters Corporation, Milford, MA, USA). The capillary voltage was set at 1.5 kV. Desolvation was achieved at 450 C with a gas flow rate of 1000 L/h. The MRMs were acquired in negative mode. The gradient and monitored MRM transitions are illustrated in Table 1.

2.5 | Validation

Validation was performed in accordance with established laboratory procedures. Briefly, selectivity was assessed by using 10 different blank urines, extracted and analyzed un-spiked and spiked with both isomers at 25 ng/mL. Resolution of the two isomers as well as reproducibility of quantitation was evaluated in these samples. Limit of identification (LoI) was assessed by spiking blank urine with decreasing concentrations of both isomers and determined as the lowest concentration where all samples met criteria listed in the WADA IDCR technical document. Carryover was evaluated by injecting a sample spiked at 500 ng/mL for both isomers, followed by a blank sample. Limit of quantitation (LoQ) was determined as the lowest concentration where measured concentrations were within 10% of the nominal concentration.

3 | RESULTS AND DISCUSSION

Under the assay conditions described above, the signal intensity for Δ^8-THC-COOH was lower than the intensity for an identical concentration of Δ^9-THC-COOH. However, both isomers could be reproducibly resolved over a range of 0.5 to 500 ng/mL. See Figure 1. The LoI was established as 0.5 ng/mL for both isomers.

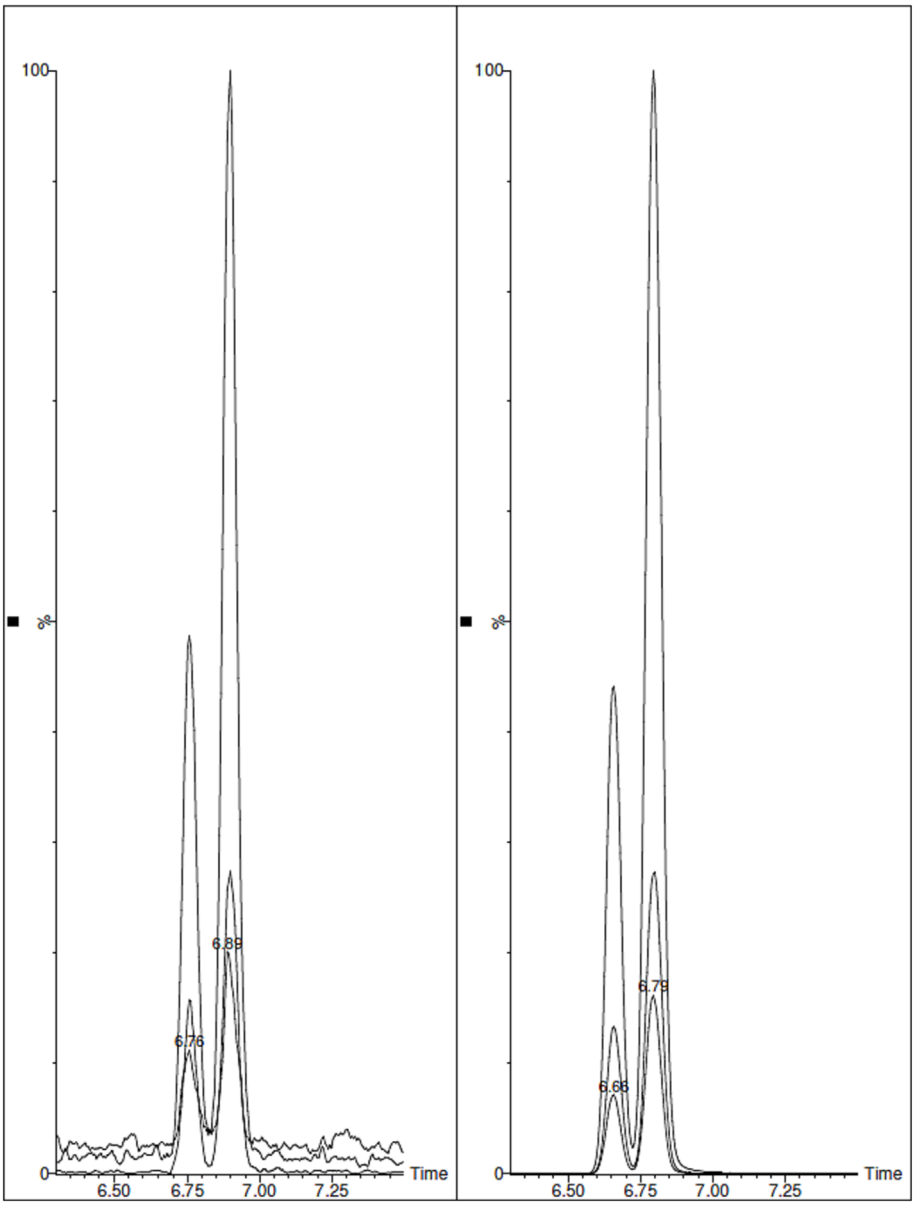

FIGURE 1 LC–MS/MS chromatograms for spiked QC samples containing 0.5 ng/mL (left panel) and 500 ng/mL (right panel) of both THC-COOH isomers. The Δ^8 isomer is the earlier eluting analyte. The most abundant MRM transition (343.2 -> 299) was used as the quantifier signal. Two other transitions (343.2 -> 245.2 and 343.2 -> 325.2) were monitored as qualifiers.

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3631 by Ohio State University Ohio Sta, Wiley Online Library on [16/01/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

LoQ was 2 ng/mL for Δ^8-THC-COOH and 0.5 ng/mL for Δ^9-THC-COOH.

The GC–MS/MS method used in the laboratory for initial screening does not resolve the two isomers. The screen estimate is therefore a measure of total THC COOH content. An example is illustrated in Figure 2. The GC–MS/MS screening produced a single peak, whereas the corresponding LC extract, when analyzed on the optimized confirmation method, resolved the Δ^8 and Δ^9 isomers. One thousand samples with an initial GC–MS/MS screening concentration between 5 and 450 ng/mL were reanalyzed by LC–MS/MS. Of these, 119 showed the presence of Δ^8-THC-COOH. The relative amount of the Δ^8 isomer in these samples ranged from 5% to 100%. See Figure 3. Separation of the two isomers allows quantitation of each isomer; however, the rules regarding reporting of the Δ^8-THC-COOH isomer are unclear.

Guidelines issued by the Substance Abuse and Mental Health Services Administration (SAMHSA) for workplace drug testing in the United States specify detection of Δ^9-THC-COOH for detection of cannabis use at a confirmatory threshold of 15 ng/mL. For World Anti-Doping Agency (WADA) code signatories, cannabinoids (except CBD) are prohibited in competition.[14] For samples collection in-competition, Δ^9-THC-COOH has a threshold of 150 ng/mL in urine and a reporting decision limit of 180 ng/mL.[15] On the other hand, synthetic cannabimimetics have a minimum reporting limit of 1 ng/mL.[16] Professional sporting organizations that are not WADA code signatories have thresholds that fall between 5 and 150 ng/mL; however the status of Δ^8-THC-COOH remains unclear.

While Δ^8 and Δ^9-THC appear to be pharmacologically similar, there are contradictory reports about their binding affinity to cannabinoid receptors. Huffman et al. reported similar binding affinities for both molecules at the CB1 and CB2 receptors,[17] whereas Radwan et al. reported lower binding affinity at the CB1 receptor and higher binding affinity at the CB2 receptor for Δ^8-THC relative to Δ^9-THC.[18] Nevertheless, Δ^8-THC has been postulated to be less psychoactive that Δ^9-THC.[19] Evaluation of impairment and kinetic patterns after controlled administration of Δ^8-THC containing products seem to indicate that Δ^8-THC is indeed less psychoactive than Δ^9-THC even though the elimination profiles for both isomers show similar patterns.[20] An online survey of 521 participants across 38 states in the United States also appeared to indicate lower psychoactive effects for Δ^8-THC.[21] On average, the participants reported that the experiences with Δ^8-THC were characterized by relaxation, pain relief, and

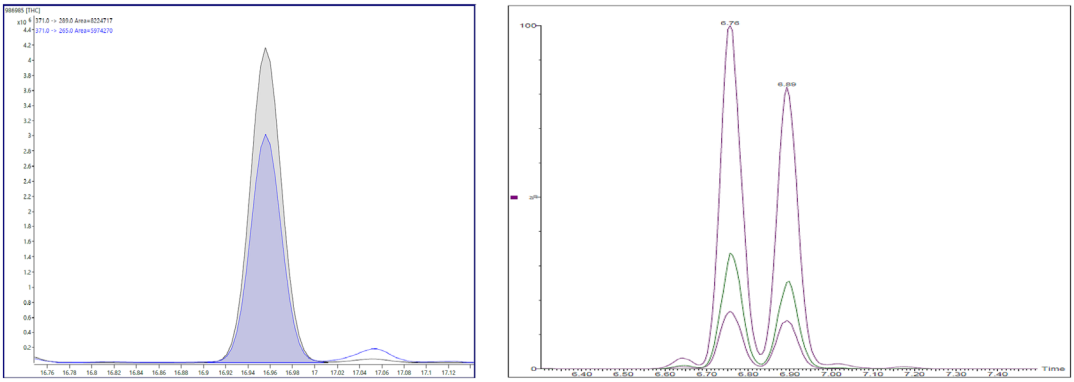

FIGURE 2 Left panel shows chromatogram from GC–MS/MS screening that does not resolve the Δ^8 and Δ^9 isomers. Right panel shows reinjection of the corresponding LC screening fraction on a method that resolves the two isomers.

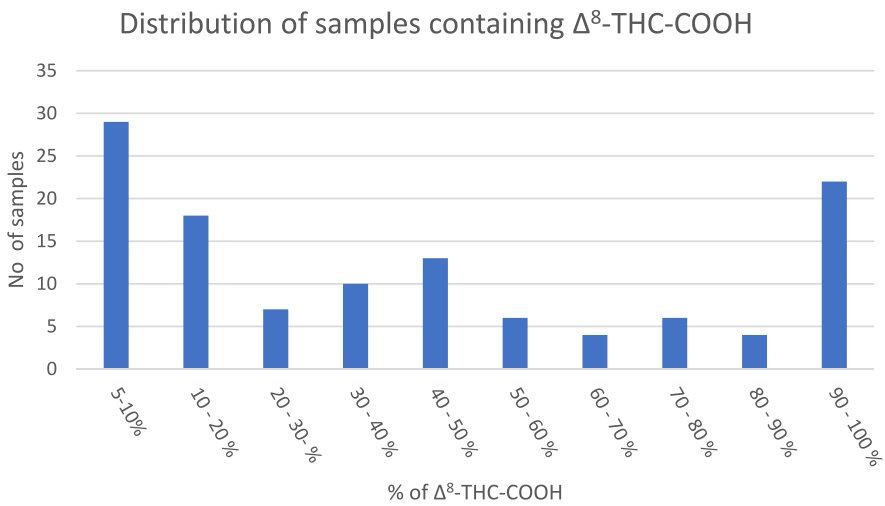

FIGURE 3 Distribution of samples containing detectable amounts of Δ^8-THC-COOH. This distribution represents the 12% of samples where Δ^8-THC-COOH was detectable. The X-axis indicates the ratio of Δ^8-THC-COOH to the sum of Δ^8 and Δ^9 isomers, expressed as a percentage.

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3631 by Ohio State University Ohio Sta, Wiley Online Library on [16/01/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

euphoria but were less intense and of shorter duration than with Δ^9-THC. Since the pharmacological effect appears very similar to Δ^9-THC, a case could be made for application of the current WADA threshold to the sum of both THC COOH isomers as opposed to exclusively to Δ^9-THC COOH.

4 | CONCLUSIONS

The legal grey-area status of Δ^8-THC has led to an increase in its market availability. Further, some manufacturers claim unproven medicinal benefits to the use of Δ^8-THC. The above factors have combined to cause an increase in the use of Δ^8-THC products in the athlete population. Since this study involves a large sample size of athletes from around the Unites States and includes multiple sporting disciplines and federations, the study population may also provide some insight into the abuse of Δ^8-THC in the general population. The potential for abuse or side effects is outside the scope of this study; however it does appear that resolution of the two isomers is necessary to comply with the thresholds in antidoping or workplace drug testing programs, which are specific to Δ^9-THC COOH. Athletes, colleges, and universities, as well as professional sports organizations, may want to educate themselves about the prevalence of Δ^8-THC use among athletes and the risks associated with this use.

ACKNOWLEDGMENTS

Funding for the study was provided by the Sports Medicine Research and Testing Laboratory.

CONFLICT OF INTEREST STATEMENT

None of the authors declare any conflicts of interest.

ORCID

Vinod S. Nair https://orcid.org/0000-0003-0093-8093

REFERENCES

1. Elsohly MA, Slade D. Chemical constituents of marijuana: the complex mixture of natural cannabinoids. *Life Sci*. 2005;78(5):539-548. doi:10.1016/j.lfs.2005.09.011
2. Gulck T, Moller BL. Phytocannabinoids: origins and biosynthesis. *Trends Plant Sci*. 2020;25(10):985-1004. doi:10.1016/j.tplants.2020.05.005
3. Hanus LO. Pharmacological and therapeutic secrets of plant and brain (endo)cannabinoids. *Med Res Rev*. 2009;29(2):213-271. doi:10.1002/med.20135
4. Hazekamp A, Fischedick JT. Cannabis—from cultivar to chemovar. *Drug Test Anal*. 2012;4(7-8):660-667. doi:10.1002/dta.407
5. Babalonis S, Raup-Konsavage WM, Akpunonu PD, Balla A, Vrana KE. Delta(8)-THC: legal status, widespread availability, and safety concerns. *Cannabis Cannabinoid Res*. 2021;6(5):362-365. doi:10.1089/can.2021.0097
6. Kruger DJ, Kruger JS. Consumer experiences with delta-8-THC: medical use, pharmaceutical substitution, and comparisons with delta-9-THC. *Cannabis Cannabinoid Res*. 2023;8(1):166-173. doi:10.1089/can.2021.0124
7. Kiselak TD, Koerber R, Verbeck GF. Synthetic route sourcing of illicit at home cannabidiol (CBD) isomerization to psychoactive cannabinoids using ion mobility-coupled-LC-MS/MS. *Forensic Sci Int*. 2020;308:110173. doi:10.1016/j.forsciint.2020.110173
8. Erickson B. In: News CE, ed. *Delta-8-THC Craze Concerns Chemists*. Vol.99; 2021.
9. Administration, U.F.a.D. *5 things to know about delta-8 tetrahydrocannabinol—delta-8 THC*. (2022).
10. Centers for Disease Control and Prevention, T. Increases in availability of cannabis products containing delta-8 THC and reported cases of adverse events. In: *Health Alert Network (HAN)*. Centers for Disease Control and Prevention; 2021.
11. Helander A, Johansson M, Andersson A, Villen T. Analytical and medico-legal problems linked to the presence of delta-8-tetrahydrocannabinol (delta-8-THC): results from urine drug testing in Sweden. *Drug Test Anal*. 2022;14(2):371-376. doi:10.1002/dta.3190
12. Rzeppa S, Große J, Rautenberg C, et al. Emergence of the less common cannabinoid delta(8)-tetrahydrocannabinol in a doping sample. *Drug Test Anal*. 2021;13(11-12):1936-1943. doi:10.1002/dta.3159
13. Miller GD, Nair V, Morrison MS, Summers M, Willick SE, Eichner D. Intranasal delivery of Natesto(R) testosterone gel and its effects on doping markers. *Drug Test Anal*. 2016;8(11-12):1197-1203. doi:10.1002/dta.2106
14. World Anti-Doping Agency, T. *The 2022 Prohibited List*. International Standard; 2022.
15. World Anti-Doping Agency, T. WADA technical document—TD2022DL: decision limits for the confirmatory quantification of exogenous threshold substances by chromatography-based analytical methods. (2022).
16. World Anti-Doping Agency, T. WADA technical document—TD2022MRPL: minimum required performance levels and applicable minimum reporting levels for non-threshold substances analyzed by chromatographic—mass spectrometric analytical methods. (2022).
17. Huffman JW, Liddle J, Yu S, et al. 3-(1′,1′-dimethylbutyl)-1-deoxy-delta8-THC and related compounds: synthesis of selective ligands for the CB2 receptor. *Bioorg Med Chem*. 1999;7(12):2905-2914. doi:10.1016/S0968-0896(99)00219-9
18. Radwan MM, ElSohly MA, el-Alfy AT, et al. Isolation and pharmacological evaluation of minor cannabinoids from high-potency *Cannabis sativa*. *J Nat Prod*. 2015;78(6):1271-1276. doi:10.1021/acs.jnatprod.5b00065
19. Razdan RK. Structure-activity relationships in cannabinoids. *Pharmacol Rev*. 1986;38(2):75-149.
20. Wurz GT, Montoya E, DeGregorio MW. Examining impairment and kinetic patterns associated with recent use of hemp-derived delta(8)-tetrahydrocannabinol: case studies. *J Cannabis Res*. 2022;4(1):36. doi:10.1186/s42238-022-00146-9
21. Kruger JS, Kruger DJ. Delta-8-THC: delta-9-THC's nicer younger sibling? *J Cannabis Res*. 2022;4(1):4. doi:10.1186/s42238-021-00115-8

How to cite this article: Nair VS, Heybroek M, Boyle E, et al. Prevalence of carboxy-Δ^8-tetrahydrocannabiniol in antidoping samples. *Drug Test Anal*. 2024;1-5. doi:10.1002/dta.3631

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3631 by Ohio State University Ohio Sta, Wiley Online Library on [16/01/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License